D06 · Carrier Management & TMS

04 · Real-time shipment visibility & supply chain control tower

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Shipment visibility tracks freight in real time across carriers and modes; a logistics control tower adds alert thresholds, owners and escalation rules so exceptions are detected and resolved before the customer is affected.

2 WHY IT MATTERS

Visibility alone does not change outcomes; **action does**. Many control towers become expensive dashboards that confirm problems after customers have already called. The value comes from closed-loop exception management — detect, assign, act, inform, learn — and from feeding recurring root causes back to carrier scorecards and planning.

3 HOW IT IS DONE

1. **Connect the data sources.** TMS, carrier GPS/ELD feeds, ocean and air milestones, and customer delivery windows in one event stream.
2. **Design alerts for action.** Tiered thresholds with a named owner and a defined next step; no alert without an owner.
3. **Communicate proactively.** Tell the customer about a delay before they ask, with a new ETA.
4. **Close the loop.** Log root causes, review them monthly and push fixes into carriers, routes and planning parameters.

IN PRACTICE

Design alerts for action, not for information

Every alert should have a clear owner and a defined next step. Alerts that no one is expected to act on should not exist.

Close the loop into carrier and process improvement

Repeated exceptions with the same root cause must feed the carrier scorecard and internal process reviews.

Proactive customer communication is a core tower output

A customer who is told about a delay before they have to ask experiences a different level of service.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case — Mid-sized manufacturer implementing a control tower — Previously learned about delays only when the customer called.

Metric	Before	After 9 months
% of delays detected before customer contact	12%	81%
Average time from delay detection to customer notification	N/A (reactive)	4.2 hours
Customer complaints related to late delivery	Baseline	-34%
Expedite freight triggered by late discovery	Baseline	-27%

Result: The largest gain was not operational cost reduction but the shift from reactive to proactive customer communication, which measurably improved perceived reliability.

HOW IT IS MEASURED

Proactive Detection Rate %

(Delays detected by the tower before customer contact / Total delays) × 100

Benchmark: >75% is a solid operating target

Mean Time to Detect / Mean Time to Resolve

Average hours from event occurrence to detection and from detection to resolution

Benchmark: Track both; detection without resolution creates little value

WHAT YOU WOULD USE

Visibility & Control Tower

project44 / FourKites / Shippeo — Leading multimodal visibility platforms with predictive ETAs and exception management.

SAP Logistics Control Tower / Blue Yonder Luminate — Control tower capabilities integrated with broader planning suites.

4 SO WHAT

Build the tower around decisions, not screens. Unify the data, give every alert an owner, warn customers early and turn recurring exceptions into fixes upstream.

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